

Principle Safety References

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0. Preface

These operating instructions are designed to familiarize the user with the machine/plant and its designated use.

The instruction manual contains important information on how to operate the machine/plant safely, properly and most efficiently. Observing these instructions helps to avoid danger, to reduce repair costs and downtimes and to increase the reliability and life of the machine/plant.

The instruction manual is to be supplemented by the respective national rules and regulations for accident prevention and environmental protection.

The operating instructions must always be available wherever the machine/plant is in use.

These operating instructions must be read and applied by any person in charge of carrying out work with and on the machine/plant, such as

- operation including setting up, troubleshooting in the course of work, evacuation of production waste, care and disposal of fuels and consumables.
- maintenance (servicing, inspection, repair) and/or
- transport.

In addition to the operating instructions and to the mandatory rules and regulations for accident prevention and environmental protection in the country and place of use of the machine/plant, the generally recognized technical rules for safe and proper working must also be observed.

1.0 Principle Safety References

1.1 Warning Signs and Symbols

The following signs and designations are used in the manual to designate instructions of particular importance.

Reference:

Important particular indication to the economical operation of the machine/plant.

Attention !

Attention:

refers to special information and/or orders and prohibitions directed towards preventing damage.



Danger:

refers to orders and prohibitions designed prevent injury or extensive damage.

Please note:

Used signs, similar to traffic signs, are self-explanatory.

1.2 Basic Operation and Designated Use of the Machine/Plant

1.2.1 The machine/plant has been built in accordance with state-of-the-art standards and the recognized safety rules. Nevertheless, its use may constitute a risk to life and limb of the user or of third parties, or cause damage to the machine and to other material property.

1.2.2 The machine/plant must only be used in technically perfect condition in accordance with its designated use and the instructions set out in the operating manual, and only by safety-conscious persons who are fully aware of the risks involved in operating the machine/plant. Any functional disorders, especially those affecting the safety of the machine/plant, should therefore be rectified immediately.

1.2.3 The machine/plant is designed exclusively for ^{**}). Using the machine/plant for purposes other than those mentioned above (such as for ^{**}) is considered contrary to its designated use. The manufacturer/supplier cannot be held liable for any damage resulting from such use. The risk of such misuse lies entirely with the user.

Operating the machine within the limits of its designated use also involves observing the instructions set out in the operating manual and complying with the inspection and maintenance directives.

1.3 Organizational Measures

1.3.1 The operating instructions must always be at hand at the place of use of the machine/plant, e.g. by stowing them in the tool compartment or tool-box provided for such purpose.

1.3.2 In addition to the operating instructions, observe and instruct the user in all other generally applicable legal and other mandatory regulations relevant to accident prevention and environmental protection.

These compulsory regulations may also deal with the handling of hazardous substances, issuing and/or wearing of personal protective equipment, or traffic regulations.

1.3.3 The operating instructions must be supplemented by instructions covering the duties involved in supervising and notifying special organizational features, such as job organization, working sequences or the personnel entrusted with the work.

1.3.4 Personnel entrusted with work on the machine must have read the operating instructions and in particular the chapter on safety before beginning work. Reading the instructions after work has begun is too late. This applies especially to persons working only occasionally on the machine, e. g. during setting up or maintenance.

^{**}) see specific Operating Instructions (BA)

- 1.3.5 Check - at least from time to time - whether the personnel is carrying out the work in compliance with the operating instructions and paying attention to risks and safety factors.
- 1.3.6 For reasons of security, long hair must be tied back or otherwise secured, garments must be close-fitting and no jewellery - such as rings - may be worn. Injury may result from being caught up in the machinery or from rings catching on moving parts.
- 1.3.7 Use protective equipment wherever required by the circumstances or by law.
- 1.3.8 Observe all safety instructions and warnings attached to the machine/plant.
- 1.3.9 See to it that safety instructions and warnings attached to the machine are always complete and perfectly legible.
- 1.3.10 In the event of safety-relevant modifications or changes in the behaviour of the machine/plant during operation, stop the machine/plant immediately and report the malfunction to the competent authority/ person.
- 1.3.11 Never make any modifications, additions or conversions which might affect safety without the supplier's approval. This also applies to the installation and adjustment of safety devices and valves as well as to welding work on load-bearing elements.
- 1.3.12 Spare parts must comply with the technical requirements specified by the manufacturer. Spare parts from original equipment manufacturers can be relied to do so.
- 1.3.13 Never modify the software of programmable control systems.
- 1.3.14 Replace hydraulic hoses within stipulated and appropriate intervals even if no safety-relevant defects have been detected.
- 1.3.15 Adhere to prescribed intervals or those specified in the operating instructions for routine checks and inspections.
- 1.3.16 For the execution of maintenance work, tools and workshop equipment adapted to the task on hand are absolutely indispensable.
- 1.3.17 The personnel must be familiar with the location and operation of fire extinguishers.
- 1.3.18 Observe all fire-warning and fire-fighting procedures.
- 1.4 Selection and Qualification of Personnel - Basic Responsibilities**
- 1.4.1 Any work on and with the machine/plant must be executed by reliable personnel only. Statutory minimum age limits must be observed.

- 1.4.2 Employ only trained or instructed staff and set out clearly the individual responsibilities of the personnel for operation, set-up, maintenance and repair.
- 1.4.3 Make sure that only authorized personnel works on or with the machine.
- 1.4.4 Define the machine operator's responsibilities - also with regard to observing traffic regulations - giving the operator the authority to refuse instructions by third parties that are contrary to safety.
- 1.4.5 Do not allow persons to be trained or instructed or persons taking part in a general training course to work on or with the machine/plant without being permanently supervised by an experienced person.
- 1.4.6 Work on the electrical system and equipment of the machine/plant must be carried out only by a skilled electrician or by instructed persons under the supervision and guidance of a skilled electrician and in accordance with electrical engineering rules and regulations.
- 1.4.7 Work on gas-fuelled equipment (gas consumers) may be carried out by specially trained personnel only.
- 1.4.8 Work on the hydraulic system must be carried out only by personnel with special knowledge and experience of hydraulic equipment.
- 1.5 Safety Instructions Governing specific Operational Phases**
- 1.5.1 Standard operation
- 1.5.1.1 Avoid any operational mode that might be prejudicial to safety.
- 1.5.1.2 Take the necessary precautions to ensure that the machine is used only when in a safe and reliable state! Operate the machine only if all protective and safety-oriented devices, such as removable safety devices, emergency shut-off equipment, sound-proofing elements and exhausters, are in place and fully functional.
- 1.5.1.3 Check the machine/plant at least once per working shift for obvious damage and defects. Report any changes (incl. changes in the machine's working behaviour) to the competent organization/person immediately. If necessary, stop the machine immediately and lock it.
- 1.5.1.4 In the event of malfunctions, stop the machine/plant immediately and lock it. Have any defects rectified immediately.
- 1.5.1.5 During start-up and shut-down procedures always watch the indicators in accordance with the operating instructions.
- 1.5.1.6 Before starting up or setting the machine/plant in motion, make sure that nobody is at risk.
- 1.5.1.7 The selector switch must be set to "Normal" and locked.

- 1.5.1.8 Never switch off or remove suction and ventilation devices when the machine is in operation.
- 1.5.2 Special work in conjunction with utilization of the machine/plant and maintenance and repairs during operation; disposal of parts and consumables
- 1.5.2.1 Observe the adjusting, maintenance and inspection activities and intervals set out in the operating instructions, including information on the replacement of parts and equipment. These activities may be executed by skilled personnel only.
- 1.5.2.2 Brief operating personnel before beginning special operations and maintenance work, and appoint a person to supervise the activities.
- 1.5.2.3 In any work concerning the operation, conversion or adjustment of the machine and its safety-oriented devices or any work related to maintenance, inspection and repair, always observe the start-up and shut-down procedures set out in the operating instructions and the information on maintenance work.
- 1.5.2.4 Ensure that the maintenance area is adequately secured.
- 1.5.2.5 If the machine/plant is completely shut down for maintenance and repair work, it must be secured against inadvertent starting by:
- locking the principal control elements and removing the ignition key and/or
 - attaching a warning sign to the main switch.
- 1.5.2.6 To avoid the risk of accidents, individual parts and large assemblies being moved for replacement purposes should be carefully attached to lifting tackle and secured. Use only suitable and technically perfect lifting gear and suspension systems with adequate lifting capacity. Never work or stand under suspended loads.
- 1.5.2.7 The fastening of loads and the instructing of crane operators should be entrusted to experienced persons only. The marshaller giving the instructions must be within sight or sound of the operator.
- 1.5.2.8 For carrying out overhead assembly work always use specially designed or otherwise safety-oriented ladders and working platforms. Never use machine parts as a climbing aid. Wear a safety harness when carrying out maintenance work at greater heights.
- Keep all handles, steps, handrails, platforms, landings and ladders free from dirt, snow and ice.
- 1.5.2.9 Clean the machine, especially connections and threaded unions, of any traces of oil, fuel or preservatives before carrying out maintenance/repair. Never use aggressive detergents. Use lint-free cleaning rags.

- 1.5.2.10 Before cleaning the machine with water, steam jet (high-pressure cleaning) or detergents, cover or tape up all openings which - for safety and functional reasons - must be protected against water, steam or detergent penetration. Special care must be taken with electric motors and switch gear cabinets.
- 1.5.2.11 Ensure during cleaning of the machine that the temperature sensors of the fire-warning and fire-fighting systems do not come into contact with hot cleaning agents as this might activate the fire-fighting system.
- 1.5.2.12 After cleaning, remove all covers and tapes applied for that purpose.
- 1.5.2.13 After cleaning, examine all fuel, lubricant and hydraulic fluid lines for leaks, loose connections, chafe marks and damage. Any defects found must be rectified without delay.
- 1.5.2.14 Always tighten any screwed connections that have been loosened during maintenance and repair.
- 1.5.2.15 Any safety devices removed for set-up, maintenance or repair purposes must be refitted and checked immediately upon completion of the maintenance and repair work.
- 1.5.2.16 Ensure that all consumables and replaced parts are disposed of safely and with minimum environmental impact.

1.6 Warning of Special Dangers

1.6.1 Electric energy

- 1.6.1.1 Use only original fuses with the specified current rating. Switch off the machine/plant immediately if trouble occurs in the electrical system.
- 1.6.1.2 Work on the electrical system or equipment may only be carried out by a skilled electrician himself or by specially instructed personnel under the control and supervision of such electrician and in accordance with the applicable electrical engineering rules.
- 1.6.1.3 If provided for in the regulations, the power supply to parts of machines and plants, on which inspection, maintenance and repair work is to be carried out must be cut off. Before starting any work, check the de-energized parts for presence of power and ground or short-circuit them in addition to insulating adjacent live parts and elements.
- 1.6.1.4 The electrical equipment of machines/plants is to be inspected and checked at regular intervals. Defects such as loose connections or scorched cables must be rectified immediately.

1.6.1.5 Necessary work on live parts and elements must be carried out only in the presence of a second person who can cut off the power supply in case of danger by actuating the emergency shut-off or main power switch. Secure the working area with a red-and-white safety chain and a warning sign. Use insulated tools only.

1.6.1.6 Before starting work on high-voltage assemblies and after cutting out the power supply, the feeder cable must be grounded and components, such as capacitors, short-circuited with a grounding rod.

1.6.2 Gas, Dust, Steam and Smoke

1.6.2.1 Carry out welding, flame-cutting and grinding work on the machine/plant only if this has been expressly authorized, as there may be a risk of explosion and fire.

1.6.2.2 Before carrying out welding, flame-cutting and grinding operations, clean the machine/plant and its surroundings from dust and other inflammable substances and make sure that the premises are adequately ventilated (risk of explosion).

1.6.2.3 Observe any existing national regulations if work is to be carried out in narrow rooms.

1.6.3 Hydraulic and Pneumatic Equipment

1.6.3.1 Work on hydraulic equipment may be carried out only by persons having special knowledge and experience in hydraulic systems.

1.6.3.2 Check all lines, hoses and screwed connections regularly for leaks and obvious damage. Repair damage immediately. Splashed oil may cause injury and fire.

1.6.3.3 Depressurize all system sections and pressure pipes (hydraulic system, compressed-air system) to be removed in accordance with the specific instructions for the unit concerned before carrying out any repair work.

1.6.3.4 Hydraulic and compressed-air lines must be laid and fitted properly. Ensure that no connections are interchanged. The fittings, lengths and quality of the hoses must comply with the technical requirements.

1.6.4 Noise

1.6.4.1 During operation, all sound baffles must be closed.

1.6.4.2 Always wear the prescribed ear protectors.

1.6.5 Oil, Grease and other Chemical Substances

1.6.5.1 When handling oil, grease and other chemical substances, observe the product-related safety regulations.

1.6.5.2 Be careful when handling hot consumables (risk of burning or scalding).

1.7 **Mobile Machinery and Equipment**

(Machinery and equipment used at frequently changing places of operation)

1.7.1 For loading only use lifting gear and tackle of sufficient capacity.

1.7.2 Appoint a competent marshaller to assist in the lifting operations.

1.7.3 Lift machinery and equipment properly with suitable lifting gear and only in accordance with the operating instructions (fixing points for lifting tackle, etc ...).

1.7.4 Only use suitable means of transport of adequate carrying capacity.

1.7.5 Fasten the loads safely using the suitable fixing points.

1.7.6 Before or immediately after completion of the loading operations the machine/plant must be secured by means of recommended/supplied devices against unintentional changes of position and a corresponding warning sign attached to the machine/plant.

Before recommissioning the machine/plant these devices must be properly removed.

1.7.7 Carefully refit and fasten all parts to be removed for transport purposes before recommissioning the machine/plant.

1.7.8 Cut off the external power supply of the machine or plant even if only minor changes of place are envisaged. Properly reconnect the machine to the supply mains before recommissioning.

1.7.9 For recommissioning only proceed in accordance with the operating instructions.